AI-Powered Smart Gym System

User Stories - 59 stories in 9 epics, derived from the Requirements Engineering document.

Format: As a [user], I want [goal], so that [benefit]. Personas come from the document: novice gym user, experienced lifter, home trainee, nutrition-led user, plus the admin. Pri is MoSCoW taken from the source requirement. † = the source gives no priority (requirement is a target or user requirement only), so Should is an assumption. Trace lists the FR/NFR/UR IDs and the matching use case numbers from the use case document.

1. Account and authentication

ID	User story	Acceptance criteria	Pri	Trace
US-01	As a new user, I want to register with my email and a password, so that I have a personal account for my training data.	Profile document /users/{uid} is created automatically with the required fields; password of at least 8 characters accepted, 32+ not truncated; no password stored in Firestore.	Must	FR-AUTH-01, 04 / UC-01
US-02	As a returning user, I want to sign in with Google or email, so that I can get in quickly.	Both paths resolve to the same UID when the verified email matches.	Must	FR-AUTH-02 / UC-02
US-03	As a user who forgot my password, I want a reset link by email, so that I can regain access.	Link is time-limited; the response is identical whether or not the address exists.	Must	FR-AUTH-03 / UC-03
US-04	As a user, I want to stay signed in between app launches, so that I do not log in before every workout.	Session survives restart; token refreshed silently; re-authentication only if the token is revoked or expired.	Must	FR-AUTH-05 / UC-04
US-05	As a user on a shared phone, I want sign-out to wipe my data from the device, so that others cannot see it.	Local cache of PII, biometric samples and chat history cleared; exercise catalogue kept.	Should	FR-AUTH-06 / UC-05
US-06	As a user, I want to delete my account permanently, so that my data is erased when I leave.	/users/{uid} tree and the Firebase Auth record are removed.	Should	FR-AUTH-07 / UC-06

2. Profile, onboarding and goals

ID	User story	Acceptance criteria	Pri	Trace
US-07	As a new user, I want a short onboarding, so that the app is personalised without a long form.	8 attributes (age, sex, height, weight, experience, goal, training days, activity level) collected in 8 screens or fewer.	Must	FR-PROF-01 / UC-07
US-08	As a user, I want to see my BMI, daily calories and macro targets, so that I know what to aim for.	BMI refreshes on every height/weight change; REE by Mifflin-St Jeor; TDEE and protein/carb/fat derived from activity factor and goal.	Must	FR-PROF-02, 03, 04 / UC-08
US-09	As a user, I want my heart-rate zones, so that I can train at the right intensity.	HRmax estimated as 208 - 0.7 x age; five zones shown.	Should	FR-PROF-05 / UC-09
US-10	As a user, I want a default weekly activity target that I can change, so that I have a realistic goal.	Default depends on activity level and goal (e.g. 150 min, 2+ strength days); user can override; stays within WHO guidance.	Should	FR-PROF-06, NFR-LEG-06 / UC-10
US-11	As a user, I want to switch between Arabic and English at any time, so that I use the app in my preferred language.	Change applies immediately, no restart.	Must	FR-PROF-07 / UC-11
US-12	As a home trainee, I want to declare that I have no gym equipment, so that my plan uses bodyweight exercises.	Plan adapts to declared equipment access (home / bodyweight only).	Should†	UR-HOME-02 / UC-12

3. Pose coaching (computer vision)

ID	User story	Acceptance criteria	Pri	Trace
US-13	As a novice, I want spoken and on-screen corrections while I lift, so that I fix my form without a coach.	Overlay within 100 ms, spoken cue within 300 ms; at most 1 cue per 2 s; plain language.	Must	FR-CV-07, NFR-PERF-02, UR-NOV-01 / UC-15
US-14	As a user, I want to start a tracked set in a few taps, so that I do not fiddle with my phone mid-workout.	Set starts from the home screen in 3 taps or fewer.	Must	NFR-USE-03, FR-CV-01 / UC-13
US-15	As a user, I want a countdown before scoring starts, so that I can step back and get into position.	Framing calibration with a 3-second audible countdown precedes scoring.	Must	FR-CV-10 / UC-14
US-16	As an experienced lifter, I want reps counted automatically, so that I can focus on the lift.	Rep counting at least 95% accurate on squat, deadlift and bench press vs labelled ground truth.	Must	FR-CV-05, NFR-DQ-03 / UC-16
US-17	As a user, I want a colour-coded skeleton on the camera view, so that I see at a glance if my form is good.	Green / amber / red overlay drawn on the live preview.	Must	FR-CV-08 / UC-15
US-18	As a user, I want to be told when my body is not fully visible, so that scoring is not wrong.	Scoring suspends and a reposition prompt shows if required landmarks are missing for over 1 s.	Must	FR-CV-09 / UC-17
US-19	As a user, I want to know common faults are detected, so that I avoid injury.	Rules cover knee valgus, depth/ROM, lumbar flexion, forward lean, elbow flare, asymmetry; fault classification at least 90% accurate.	Must	FR-CV-06, NFR-DQ-03 / UC-15
US-20	As a privacy-conscious user, I want my video to stay on my phone, so that nobody sees me training.	No frame, image, video or per-frame landmark is transmitted or stored; only derived session summaries.	Must	FR-CV-11, NFR-SEC-04 / UC-13
US-21	As a user, I want a summary after each set, so that I can see how I did.	Shows reps, time under tension, ROM, tempo, form score 0-100 and fault counts.	Must	FR-CV-14 / UC-19
US-22	As a user with an older phone, I want the app to adapt to my device, so that coaching stays smooth.	Lite/Full/Heavy model auto-selected by device tier with manual override; auto-downgrade if FPS stays below the floor for over 3 s.	Must	FR-CV-03, 13 / UC-18, 20
US-23	As a home trainee, I want squat, deadlift and bench press supported, so that I get useful coaching from a propped phone.	Library of at least 3 exercises in v1; works with a phone propped against a wall.	Must	FR-CV-12, UR-HOME-01 / UC-13

4. Workout session and safety

ID	User story	Acceptance criteria	Pri	Trace
US-24	As a lifter, I want to build a workout of exercises, sets, reps and load, so that I follow a plan.	Ordered exercise list; per-set completion tracked.	Must	FR-ALRT-01 / UC-21, 22
US-25	As a user, I want an automatic rest timer between sets, so that I rest the right amount.	Timer starts after each completed set.	Must	FR-ALRT-02 / UC-23
US-26	As a beginner, I want a warm-up tip and a safety notice at first use, so that I start safely.	Warm-up recommendation before the first working set; mandatory acknowledgement in the first session.	Should	FR-ALRT-04, UR-NOV-02 / UC-24
US-27	As a user, I want a warning when my heart rate is too high, so that I can ease off.	Caution shown above 90% of estimated HRmax with advice to reduce intensity.	Should	FR-ALRT-03 / UC-25
US-28	As a lifter in a gym with poor signal, I want to log workouts offline, so that I never lose data.	Coaching, set logging, rest timer and food logging work offline; writes queued and synced on reconnect with zero loss.	Must	FR-DATA-03, NFR-REL-02, UR-EXP-02 / UC-26

5. Wristband and biometrics

ID	User story	Acceptance criteria	Pri	Trace
US-29	As a user, I want to pair my wristband over Bluetooth, so that my heart rate and activity are tracked.	BLE 4.2+ link to the ESP32 band; band receives sex, age and weight from the phone.	Should†	C-TECH-04, NFR-COMP-04 / UC-27, 30
US-30	As a user, I want live heart rate, steps, calories and battery, so that I can watch my effort.	Band notifies every 2 s; UI shows each update within 200 ms of receipt.	Should†	NFR-PERF-04 / UC-28
US-31	As a user, I want the band to reconnect by itself, so that I do not re-pair mid-workout.	Link re-establishes within 10 s of returning to range.	Should†	NFR-REL-04 / UC-29
US-32	As a user without a band, I want the app to still work, so that I can use coaching and nutrition.	Missing or failed band disables only biometric features.	Must	NFR-REL-09, NFR-COMP-04 / UC-27
US-33	As a user, I want trustworthy heart-rate readings, so that zones and warnings are meaningful.	Within 10 bpm of a chest-strap reference for at least 90% of paired samples.	Should†	NFR-DQ-01 / UC-28
US-34	As a user, I want the band to last my whole session and protect its battery, so that it is dependable.	At least 8 h continuous use; deep sleep at or below 2.95 V for 5 consecutive readings.	Should†	NFR-PWR-01, 06 / UC-32
US-35	As a user, I want biometric data stored efficiently and privately, so that sync stays fast and within limits.	One aggregated document per 60 s window; retained 12 months then aggregates only.	Must	FR-DATA-04, NFR-SEC-07 / UC-31

6. Nutrition and meal planning

ID	User story	Acceptance criteria	Pri	Trace
US-36	As a nutrition-led user, I want to search Egyptian dishes in Arabic or English, so that I find foods like ful and koshary.	Fuzzy search accepts Arabic, English and transliterated spellings; results within 500 ms, offline.	Must	FR-NUT-04, UR-NUT-01 / UC-33
US-37	As a user, I want to log what I eat quickly, so that I see my daily totals.	Log by search, portion or repeat-yesterday; running daily totals.	Must	FR-NUT-07 / UC-34
US-38	As a user, I want a daily meal plan from Egyptian foods, so that I hit my targets.	3 meals plus up to 2 snacks; TDEE within 5%, protein within 10%.	Must	FR-NUT-05, UR-NUT-02 / UC-35
US-39	As a budget-conscious user, I want the plan priced in EGP with a budget limit, so that I can afford it.	Estimated cost shown; regeneration under a daily ceiling; prices older than 90 days marked stale.	Should	FR-NUT-03, 06 / UC-36
US-40	As a user with dietary restrictions, I want excluded foods never in my plan, so that I can trust it.	Exclusions recorded; no excluded item ever appears in a generated plan.	Must	FR-NUT-09, UR-NUT-03 / UC-37
US-41	As a fasting user, I want a Ramadan mode, so that meals fit iftar and suhoor.	User-defined eating window with a Ramadan preset.	Could	FR-NUT-08 / UC-38

7. Chat assistant

ID	User story	Acceptance criteria	Pri	Trace
US-42	As an Arabic speaker, I want to chat in MSA, Egyptian Arabic or English, so that it feels natural.	Replies in the language of my message.	Must	FR-CHAT-01, UR-NOV-03 / UC-39
US-43	As a user, I want answers tailored to me, so that advice fits my goal.	Goal, experience, limitations, recent workout summary and macro targets injected; raw biometric series never sent.	Must	FR-CHAT-04 / UC-39
US-44	As a user, I want the assistant to stay on fitness and nutrition and not diagnose, so that I am not misled.	Out-of-domain requests declined politely; no diagnosis or prescription; persistent non-diagnostic disclaimer.	Must	FR-CHAT-03, 06, NFR-LEG-05 / UC-43
US-45	As a user, I want fast, reliable replies, so that chat does not stall.	Reply streams (first token within 1.5 s p95); failover to secondary provider on error, timeout or 5 s without first token.	Should	FR-CHAT-02, 05, NFR-REL-05 / UC-40
US-46	As a user, I want some answer even when the AI is down, so that I am not left with nothing.	Templated fallback for common intents if both providers fail.	Should	FR-CHAT-08 / UC-41
US-47	As a user, I want to clear my chat history, so that I control what is stored.	History persisted per user and deletable on request.	Must	FR-CHAT-07 / UC-42

8. Progress analytics

ID	User story	Acceptance criteria	Pri	Trace
US-48	As a lifter, I want to see my form score over time, so that I know I am improving.	Chart of mean form score per exercise.	Should	FR-ANL-03 / UC-44
US-49	As a user, I want heart-rate trends, so that I can see fitness gains.	Per-session HR and resting HR over time from stored aggregates.	Should	FR-ANL-04 / UC-45
US-50	As a user, I want to see if I am sticking to my plan, so that I stay motivated.	Sessions completed vs planned; minutes vs weekly target.	Should	FR-ANL-05 / UC-46
US-51	As an experienced lifter, I want volume and 1RM trends, so that I can manage progressive overload.	Load, volume and estimated 1RM charts. No FR exists yet in the source (gap).	Should†	UR-EXP / UC-47

9. Localisation, privacy and administration

ID	User story	Acceptance criteria	Pri	Trace
US-52	As an Arabic user, I want a fully mirrored right-to-left UI, so that it feels native.	Complete Arabic UI; nav, icons, charts and lists mirrored; judged natural by 4 or more native reviewers.	Must	FR-LOC-01, 02, NFR-USE-06 / UC-11
US-53	As an Egyptian user, I want local formats, so that numbers make sense to me.	EGP currency, metric units, Egyptian date/time and chosen numeral system.	Must	FR-LOC-03 / UC-11
US-54	As a user, I want clear error messages, so that I know what to do.	Localised, actionable text; no raw exceptions or stack traces.	Must	FR-LOC-05
US-55	As a gym-goer, I want Arabic voice cues in Egyptian phrasing and a visual for each, so that I understand in a noisy gym.	Spoken cues in both languages; every audio cue has a visual counterpart.	Should	FR-LOC-04, NFR-USE-07 / UC-15
US-56	As a user, I want control over my personal data, so that I trust the app with health data.	Explicit per-purpose consent; access, correct, erase, object, withdraw; biometrics leave the device only as aggregates after consent.	Must	NFR-LEG-02, 03, NFR-SEC-04 / UC-51
US-57	As a user, I want to be told this is a wellness aid, so that I do not treat it as medical advice.	Statement shown at onboarding and in settings.	Must	NFR-LEG-01 / UC-51
US-58	As an admin, I want to manage food and exercise catalogues with an audit trail, so that changes are accountable.	Catalogues read-all / admin-write; every admin action written to an append-only log users cannot edit.	Must	FR-DATA-06, NFR-SEC-12 / UC-48, 49
US-59	As an admin, I want to publish a new catalogue version, so that devices refresh their offline copy.	Clients mirror catalogues at launch and refresh on version mismatch.	Should	FR-DATA-07 / UC-50